



Apple Heart Rates

Apple patents an integrated heart rate monitor for smartphones with hover touch sensors: <http://dgit.in/appleHRM>



Lava 402+ Launched

Lava launched the new budget smartphone that boasts of 3G connectivity and Android 4.2: <http://dgit.in/lava402>

Feature

lighting and high-resolution gaming. Till then they offer next to no advantage.

AMD 0 - NVIDIA 0

Stream processors

The R9 290X has 2816 cores while the NVIDIA card has 2880 cuda cores. The NVIDIA card may have more cores but the AMD card has a better implementation because of which the OpenCL performance on AMD cards is way greater than that on the NVIDIA card. The aspect is demonstrated in our OpenCL benchmark so we aren't declaring any winners here.

AMD 0 - NVIDIA 0

Texture units

The NVIDIA card has a clear advantage here with 240 texture mapping units against 176 units on the AMD R9 290X. This translates to 210 GTexel/s on the GTX 780 Ti while the R9 290X can output 176 GTexel/s.

AMD 0 - NVIDIA 1

ROPs

The NVIDIA card has 48 ROPs against 64 ROPs on the AMD card. These are propor-

floating point calculation. The performance difference scales similarly for Double Precision floating point calculations as well with the AMD R9 290X crunching 704 GFLOPS against 210 GFLOPS on the NVIDIA GTX 780 Ti.

AMD 1 - NVIDIA 0

Build Cooling

Both have copper bottomed plates onto which we have aluminium fins. There are no heat pipes on both cards and the fan design is also the same. A difference that we see right off the bat is that the AMD card's fins are closed off from the top while the NVIDIA card's fins are open ended. AMD's heatsink, however, has more copper on the base to go with compared to the NVIDIA heatsink. All this hasn't brought much difference to the table since the AMD card still runs hotter compared to the NVIDIA GTX 780 Ti.

AMD 0 - NVIDIA 0

Connectors

Both cards have the same set of connectors. But you'll notice that the AMD R9

12V rail to work optimally. AMD on the other hand hasn't mentioned any numbers here so we're in the dark regarding this segment.

AMD 0 - NVIDIA 0

Benchmarks

In this section we've decided the winner based on a 5% error margin. For a full list of all tests performed check the online article at <http://dgit.in/1guBd1g>

AMD 1 - NVIDIA 3

Return on Investment

Going back about five years, we had the GTX 285 which was the highest performing single GPU card and even to this day it performs well. Most games released in each year don't have exhaustive graphical requirements given that plenty of them are designed to work on consoles as well. So any card bought today will perform decently 5 years down the line.

For tax calculation purposes, we in India use anything between 20-30% to calculate depreciation in value of computer hardware. We'll be using 25% in this calculation. So both cards over a period of five years will depreciate in value by the following table. So at the end of five years you've lost ₹44999 on the GTX 780 Ti and ₹28982 on the R9 290X. Given that the performance difference between the two is barely anything, the price difference does bite NVIDIA on the rear especially since the ROI on both are pretty much the same.

AMD 1 - NVIDIA 0

ROI	NVIDIA GTX 780 Ti		AMD R9 290X	
	Value (Rupees)	Reduction (@25%)	Value (Rupees)	Reduction (@25%)
Current	₹59,000	₹14,750	₹38,000	₹9,500
After 1 Year(s)	₹44,250	₹11,062.5	₹28,500	₹7,125
After 2 Year(s)	₹33,187.5	₹8,296.875	₹21,375	₹5,343.75
After 3 Year(s)	₹24,890.63	₹6,222.656	₹16,031.25	₹4,007.813
After 4 Year(s)	₹18,667.97	₹4,666.992	₹12,023.44	₹3,005.859
After 5 Year(s)	₹14,000.98		₹9,017.578	

tional to the other elements of the card and come into play in the final stages where all the data computed separately needs to be combined to give the final output on the screen, which is why we won't be scoring cards based on the ROPs they have.

AMD 0 - NVIDIA 0

Floating point performance

The AMD hardware triumphs over the NVIDIA GTX 780 Ti when it comes to Single precision as well as Double precision floating point calculation. The AMD card can output a maximum of 5632 GFLOPS versus the NVIDIA card's 4500 GFLOPS of Single Precision

290X card doesn't support VGA output anymore so even if you managed to use an adapter(adapters won't fit given the lack of sockets for the analog pins) you won't be able to obtain a VGA output. However, keeping in mind that these are flagship cards that support 4K resolution it really doesn't make sense to use a VGA output anyways. So yet again there will be no winner in this segment as well.

AMD 0 - NVIDIA 0

Power

The NVIDIA card has a TDP of 250W and requires at least 42 Amperes on the

Winner

The final scores stand as follows with NVIDIA in the lead by one point.

AMD 5 - NVIDIA 6

This alone is not enough to decide the winner on the whole. When it comes to gaming the NVIDIA GTX 780 Ti is ahead of the AMD R9 290X by a small margin while the AMD tries to cover up for that with computational prowess. However, we're looking at desktop graphics cards here where the primary focus is on gaming performance and which is why the NVIDIA card takes the crown.